

# Customer Churn Prediction

Using Machine Learning

Decision Tree | SVM | K-Means



# Problem Overview

Customer churn means a customer leaves the company.

Predicting churn helps telecom companies target retention offers.

The goal is to classify each customer as Churn = Yes or No.

Business value:

Lower retention cost

Earlier risk detection

Better customer segmentation

# Dataset

Telco Customer Churn Dataset

Approximately 7043 customer records

21 attributes

Target variable: Churn

Main feature groups:

Demographic attributes

Service subscriptions

Contract and billing information

# Data Understanding

Analyze rows, columns, and data types

Check missing values

Study class distribution

Visualize important variables

Recommended charts:

Churn distribution

Contract vs Churn

Tenure distribution

Monthly charges distribution

# Preprocessing Pipeline

Drop customerID

Convert TotalCharges to numeric

Impute missing values

One-hot encode categorical variables

Scale numerical variables for SVM

Why preprocessing matters:

Models require numeric inputs.

Missing values can break training.

SVM is sensitive to feature scale.

# Sampling and Feature Analysis

Train/test split: 70% training and 30% testing

Stratified sampling keeps churn ratio stable

Feature importance explains which variables matter

Dimensionality is controlled by removing identifiers and analyzing importance

# Classification Models

## Decision Tree Classifier

Interpretable and easy to explain

Uses max\_depth to reduce overfitting

Provides feature importance

## Support Vector Machine

Good for classification

Uses RBF kernel for nonlinear boundaries

Requires scaling

# K-Means Clustering

Unsupervised method used to discover customer segments

Does not use Churn labels during training

Uses  $k=3$  customer groups

PCA visualization helps explain clusters



# Evaluation Metrics

Accuracy: overall correctness

Precision: correctness among predicted churn customers

Recall: ability to catch actual churn customers

F1-score: balance between precision and recall

Confusion matrix: detailed error breakdown

# Conclusion

The project implements a complete ML pipeline.

Decision Tree offers interpretability.

SVM can capture nonlinear churn patterns.

K-Means adds customer segmentation insight.

Final outputs are reproducible through the provided code.